

ai-os-new

Logic Bugs Re-Audit Report

Verification audit after bug fixes. Original: 14 findings. Remaining: 14 findings.

Severity	Count
CRITICAL	1
HIGH	5
MEDIUM	7
LOW	1
Total	14

LG-01

CRITICAL

Streaming cost over-counts using total_tokens as output tokens

Files: cost-manager.ts

Problem: In streamMessage, when total_tokens available from usage metadata, it's used as output token count. But total_tokens = input + output. Cost inflated by input portion.

Fix: Extract actual completion_tokens from usage metadata. Use that for output parameter.

LG-02

HIGH

Budget service double-counts tokens in cost calculation

Files: budget-service.ts

Problem: calculateCost(d.model, Math.round(tokens*inputWeight), tokens) uses tokens as both input (weighted) and output. 50% over-count for non-embedding models.

Fix: Split tokens: inputTokens=Math.round(tokens*inputWeight), outputTokens=tokens-inputTokens.

LG-03

HIGH

TaskQueue infinite microtask loop when throttle active

Files: TaskQueue.ts

Problem: When throttleMs>0, while-loop breaks (elapsed less than throttleMs). processNext called again, hits same break. Unbounded microtask loop.

Fix: When throttle blocks, set processing=false and schedule timeout-based retry.

LG-04

HIGH

Orchestration rate limiter compares tokens to cost limit

Files: orchestration-service.ts

Problem: isRateLimited() checks tokensUsed >= maxCostPerDay. tokensUsed is token count, maxCostPerDay is dollars. Unit mismatch.

Fix: Separate rateLimitTokens and rateLimitCosts maps. Compare each against correct limit.

LG-05

HIGH

ProviderBudget endSession decrements session count

Files: provider-budget.ts

Problem: endSession() decrements providerSessionCount. canStartSession() uses this for maxSessionsPerProvider. Session limit effectively unbounded - end old session, start new one.

Fix: Don't decrement if it represents total sessions. Or check providerActiveSessions for concurrent limit.

LG-06

HIGH

GroupManager emits previousState = newState

Files: group-manager.ts

Problem: syncKeyStatus sets p.status=status then emits previousState:p.status. Already overwritten so previousState always equals new state.

Fix: Save previous state before overwriting: const prevState=p.status; p.status=status; ...prevState:prevState.

LG-07

MEDIUM

key-analytics uses || instead of ?? for zero token counts

Files: key-analytics.ts

Problem: inputTokens = extExtra?.inputTokens || estimate - treats 0 as falsy. When inputTokens legitimately 0, fallback incorrectly estimates 30% of total.

Fix: Use nullish coalescing: extExtra?.inputTokens ?? estimate.

LG-08

MEDIUM

key-analytics hardcoded \$0.01/M cost vs pricing service

Files: key-analytics.ts

Problem: updateMetricsFromResponse hardcodes \$0.01/M tokens regardless of model. Primary recordUsage uses pricingService. Two paths produce different costs.

Fix: Use this.deps.pricingService.calculateCost() consistently.

LG-09

MEDIUM

compress-route maps tool fields by index after compression

Files: compress-route.ts

Problem: After compression, maps compressed[i] back to original[i] by index. compressed may have fewer items. Tool fields assigned to wrong messages.

Fix: Track message identity through compression instead of positional index.

LG-10	MEDIUM	Agent similarity score divided by count - max 0.33
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Files: agent-similarity-service.ts

Problem: calculateSimilarity returns score/count. Weights (0.4+0.3+0.3) sum to 1.0 but dividing by 3 gives max 0.333. Threshold of 0.3 never exceeded for moderate matches.

Fix: Return score directly. Weights already designed to sum to 1.0.

LG-11	MEDIUM	Contradiction detector min-length denominator inflates scores
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Files: contradiction-detector.ts

Problem: Overlap = shared/min(a.length,b.length). Short vs long claim gets overlap=1.0. False positive contradictions.

Fix: Use Jaccard similarity: shared/(a.length+b.length-shared).

LG-12	MEDIUM	Debate branching rollback meaningless cost estimation
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Files: debate-branching.ts

Problem: totalTokens = content.length*2, totalCost = confidence*0.001. Arbitrary formulas, no relation to actual token counting or pricing.

Fix: Use estimateTokenCount() and pricingService.calculateCost().

LG-13	MEDIUM	debate-consensus different numeric values = contradiction
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Files: debate-consensus.ts

Problem: an.val !== bn.val when same unit flags as contradiction. 'Revenue 5M' vs 'Costs 10M' flagged despite different metrics.

Fix: Require surrounding text semantic overlap before treating numeric differences as contradictions.

LG-14	LOW	key-analytics estimatedCost grows unboundedly
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Files: key-analytics.ts

Problem: ext.estimatedCost accumulates lifetime cost. Never reset. usageToday/usageMonthly reset on boundaries. Confusing for consumers expecting current-period metric.

Fix: Document as lifetime cumulative. Or reset when daily/monthly counters reset.